

Quiz 5

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (8 points) Let $\mathbf{F} = x^2\mathbf{i} - xe^{2y}\mathbf{j} + 3xyz\mathbf{k}$.

(a) Compute the curl of $\mathbf{F}$, $\nabla \times \mathbf{F}$.

$$\begin{aligned}\nabla \times \mathbf{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & -xe^{2y} & 3xyz \end{vmatrix} = \vec{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -xe^{2y} & 3xyz \end{vmatrix} - \vec{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ x^2 & 3xyz \end{vmatrix} + \vec{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ x^2 & -xe^{2y} \end{vmatrix} \\ &= \vec{i}(3xz - 0) - \vec{j}(3yz - 0) + \vec{k}(e^{2y} - 0) \\ &= 3xz\vec{i} - 3yz\vec{j} - e^{2y}\vec{k}\end{aligned}$$

(b) Compute $\nabla \cdot (\nabla \times \mathbf{F})$ based on the result from (a).

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{F}) &= \nabla \cdot (3xz, -3yz, e^{2y}) \\ &= \frac{\partial}{\partial x}(3xz) + \frac{\partial}{\partial y}(-3yz) + \frac{\partial}{\partial z}(e^{2y}) \\ &= 3z - 3z + 0 \\ &= 0\end{aligned}$$

Problem 2. (6 points) Find the first and second order Taylor polynomials, $P_1(x, y)$ and $P_2(x, y)$, for the function $f(x, y) = e^{3x-2y}$ at the point $\mathbf{a} = (0, 0)$.

$$\begin{aligned}Df &= [f_x \ f_y] = [3e^{3x-2y} \ -2e^{3x-2y}] \Rightarrow Df(0,0) = [3 \ -2] \\ \Rightarrow P_1(x, y) &= f(0,0) + Df(0,0) \begin{bmatrix} x-0 \\ y-0 \end{bmatrix} \\ &= 1 + [3 \ -2] \begin{bmatrix} x \\ y \end{bmatrix} = 1 + 3x - 2y \\ Hf &= \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 9e^{3x-2y} & -6e^{3x-2y} \\ -6e^{3x-2y} & 4e^{3x-2y} \end{bmatrix} \Rightarrow Hf(0,0) = \begin{bmatrix} 9 & -6 \\ -6 & 4 \end{bmatrix} \\ P_2(x, y) &= P_1(x, y) + \frac{1}{2} [x-0 \ y-0] Hf(0,0) \begin{bmatrix} x-0 \\ y-0 \end{bmatrix} \\ &= 1 + 3x - 2y + \frac{1}{2} [x \ y] \begin{bmatrix} 9 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= 1 + 3x - 2y + \frac{1}{2} (9x^2 - 12xy + 4y^2) \\ &= 1 + 3x - 2y + \frac{9}{2}x^2 - 6xy + 2y^2\end{aligned}$$

Problem 3. (6 points) Identify and determine the nature of the critical points of the function $f(x, y) = y^4 - 2xy^2 + x^3 - x$.

We first identify the critical points

$$Df(x, y) = [-2y^2 + 3x^2 - 1 \quad 4y^3 - 4xy] = 0$$

$$\Rightarrow \begin{cases} -2y^2 + 3x^2 - 1 = 0 \\ 4y^3 - 4xy = 0 \end{cases} \Rightarrow \begin{cases} 3x^2 - 1 = 2y^2 \\ 4y(y^2 - x) = 0 \end{cases} \Rightarrow y = 0 \text{ or } y^2 = x$$

$$y = 0 \Rightarrow 3x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

$$y^2 = x \Rightarrow 3x^2 - 1 = 2x \Rightarrow 3x^2 - 2x - 1 = 0 \Rightarrow x = 1 \text{ or } x = -\frac{1}{3}$$

$$x^2 = 1 \Rightarrow y = \pm 1$$

$$y^2 = -\frac{1}{3} \Rightarrow \text{no solution}$$

Four critical points:

$$\left(\frac{1}{\sqrt{3}}, 0\right), \left(-\frac{1}{\sqrt{3}}, 0\right), (1, 1), (1, -1)$$

$$Hf = \begin{bmatrix} 6x & -4y \\ -4y & 12y^2 - 4x \end{bmatrix}$$

$$i) \left(\frac{1}{\sqrt{3}}, 0\right): Hf = \begin{bmatrix} \frac{6}{\sqrt{3}} & 0 \\ 0 & -\frac{4}{\sqrt{3}} \end{bmatrix} \Rightarrow d_1 = \frac{6}{\sqrt{3}}, d_2 = -8 \Rightarrow \text{saddle point}$$

$$ii) \left(-\frac{1}{\sqrt{3}}, 0\right): Hf = \begin{bmatrix} -\frac{6}{\sqrt{3}} & 0 \\ 0 & \frac{4}{\sqrt{3}} \end{bmatrix} \Rightarrow d_1 = -\frac{6}{\sqrt{3}}, d_2 = -8 \Rightarrow \text{saddle point}$$

$$iii) (1, 1): Hf = \begin{bmatrix} 6 & -4 \\ -4 & 8 \end{bmatrix} \Rightarrow d_1 = 6, d_2 = 32 \Rightarrow \text{local minimum}$$

$$iv) (1, -1): Hf = \begin{bmatrix} 6 & 4 \\ 4 & 8 \end{bmatrix} \Rightarrow d_1 = 6, d_2 = 32 \Rightarrow \text{local minimum}$$